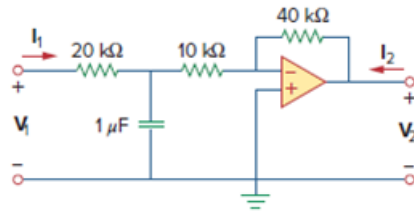


Problem

Determine the y parameters at $\omega = 1,000$ rad/s for the op amp circuit in Fig. 19.111. Find the corresponding h parameters.

Figure 19.111



Step-by-step solution

Step 1 of 4

Consider the following op amp circuit with nodes marked as 1 and 2.

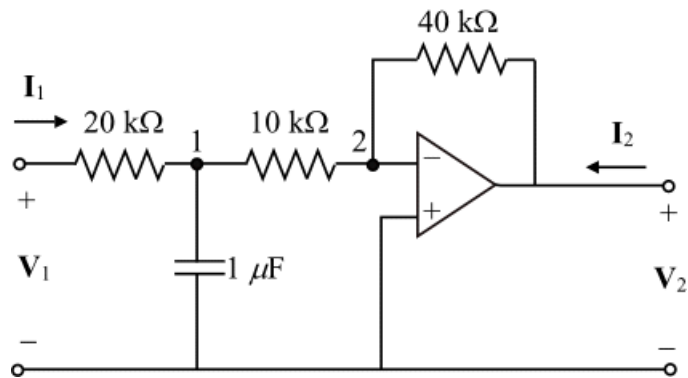


Figure 1

Step 2 of 4

Apply Kirchhoff's current law at node 1.

$$\frac{V_1 - V_x}{20 \text{ k}} = \left(\frac{1}{j\omega C} \right) + \frac{V_x}{10 \text{ k}}$$

$$\frac{V_1 - V_x}{20 \text{ k}} = \left(\frac{1}{j(1000)(10^{-6})} \right) + \frac{V_x}{10 \text{ k}}$$

$$\frac{V_1 - V_x}{20 \text{ k}} = -\frac{V_x}{j10^3} + \frac{V_x}{10 \text{ k}}$$

$$\frac{V_1}{20 \text{ k}} = \frac{V_x}{20 \text{ k}} + \frac{jV_x}{1 \text{ k}} + \frac{V_x}{10 \text{ k}}$$

$$V_1 = (3 + j20) V_x$$

Apply Kirchhoff's current law at node 2.

$$\frac{V_x - 0}{10 \text{ k}} = \frac{0 - V_2}{40 \text{ k}}$$

$$V_x = -\frac{1}{4} V_2$$

Substitute $-\frac{1}{4} V_2$ for V_x .

$$V_1 = (3 + j20) \left(-\frac{1}{4} V_2 \right)$$

$$V_1 + (0.75 + j5) V_2 = 0 \dots\dots (1)$$

Step 3 of 4

Write the expression for input current I_1 .

$$I_1 = \frac{V_1 - V_x}{20 \text{ k}}$$

Substitute $-\frac{1}{4} V_2$ for V_x .

$$I_1 = \frac{V_1 - \left(-\frac{1}{4} V_2 \right)}{20 \text{ k}}$$

$$4V_1 + V_2 = 80 \text{ k} I_1$$

$$5 \times 10^{-5} V_1 + 1.25 \times 10^{-5} V_2 = I_1 \dots\dots (2)$$

Compare equations (1) and (2) with the following y-parameter equations:

$$I_1 = y_{11} V_1 + y_{12} V_2$$

$$I_2 = y_{21} V_1 + y_{22} V_2$$

Thus,

$$y_{11} = 5 \times 10^{-5}$$

$$y_{12} = 1.25 \times 10^{-5}$$

$$y_{21} = 1$$

$$y_{22} = 0.75 + j5$$

Thus, the y-parameters of the network are $\begin{bmatrix} 5 \times 10^{-5} & 1.25 \times 10^{-5} \\ 1 & 0.75 + j5 \end{bmatrix}$.

Step 4 of 4

Use the following conversion formulae to convert y -parameters into h -parameters.

$$h_{11} = \frac{1}{y_{11}}$$

$$h_{12} = -\frac{y_{12}}{y_{11}}$$

$$h_{21} = \frac{y_{21}}{y_{11}}$$

$$h_{22} = \frac{y_{11}y_{22} - y_{12}y_{21}}{y_{11}}$$

Substitute the y -parameters values.

$$\begin{aligned} h_{11} &= \frac{1}{y_{11}} \\ &= \frac{1}{5 \times 10^{-5}} \\ &= 20,000 \end{aligned}$$

$$\begin{aligned} h_{12} &= -\frac{y_{12}}{y_{11}} \\ &= -\frac{1.25 \times 10^{-5}}{5 \times 10^{-5}} \\ &= -0.25 \end{aligned}$$

$$\begin{aligned} h_{21} &= \frac{y_{21}}{y_{11}} \\ &= \frac{1}{5 \times 10^{-5}} \\ &= 20,000 \end{aligned}$$

$$\begin{aligned} h_{22} &= \frac{y_{11}y_{22} - y_{12}y_{21}}{y_{11}} \\ &= \frac{(5 \times 10^{-5})(0.75 + j5) - (1.25 \times 10^{-5})(1)}{5 \times 10^{-5}} \\ &= (0.75 + j5) - 0.25 \\ &= 0.5 + j5 \end{aligned}$$

Thus, the h parameters of the network are $\begin{bmatrix} 20,000 \Omega & -0.25 \\ 20,000 & 0.5 + j5 \text{ S} \end{bmatrix}$.